

AP Precalculus: 2026–2027

Unit 1: Polynomial and Rational Functions Lesson 1.11: Equivalent Representations of Polynomial and Rational Expressions

Primary Objective: Students will rewrite polynomial and rational expressions in equivalent forms (factored and standard), perform polynomial long division to express the division algorithm $f = gq + r$ and identify slant asymptotes, and apply the binomial theorem via Pascal's Triangle to expand $(a + b)^n$. *Skills: MP 1.B, MP 3.B*

Priority Ideas & Skills

AP Skills

MP 1.B — Identify an appropriate mathematical rule or procedure based on the classification of a given expression.

MP 3.B — Apply an appropriate mathematical definition, theorem, or test.

Key Understandings

- Factored form reveals zeros, asymptotes, holes, and domain; standard form reveals end behavior and degree. (EK 1.11.A.1, 1.11.A.2)
- Different analytic representations answer different questions in context. (EK 1.11.A.3)
- Division algorithm: $f(x) = g(x)q(x) + r(x)$, $\deg(r) < \deg(g)$. (EK 1.11.B.1)
- The quotient q from long division gives the equation of a slant asymptote. (EK 1.11.B.2)
- Pascal's Triangle coefficients expand $(a + b)^n$. (EK 1.11.C.1)

Vocabulary, Concepts, & Theorems

factored form

Expression written as a product of linear and/or irreducible quadratic factors; directly reveals zeros, holes, and asymptotes.

standard form

Polynomial written with terms in descending order; reveals degree, leading coefficient, and end behavior.

polynomial long division

Algorithm producing $f = g \cdot q + r$ with $\deg(r) < \deg(g)$.

quotient / remainder

$q(x)$ is the quotient; $r(x)$ is the remainder in the division algorithm.

slant asymptote

A non-horizontal, non-vertical linear asymptote $y = mx + b$; occurs when $\deg(f) = \deg(g) + 1$; given by the quotient q .

binomial theorem

$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$; coefficients from Pascal's Triangle.

Pascal's Triangle

Triangular array where each entry is the sum of the two entries above; row n gives binomial coefficients for $(a + b)^n$.

Activate Prior Knowledge & Spiral Review

Spiral review (warm-up) covers: factoring polynomials (Lesson 1.5), identifying rational zeros and zeros of polynomials (Lesson 1.6/1.7), recognizing holes and VAs from shared factors (Lesson 1.10), and using leading-term analysis for end behavior (Lesson 1.8/1.9). Students factor $x^3 - x^2 - 6x$, classify a hole vs. VA, and describe end behavior from standard form.

Hook

Write three rational functions on the board:

$$\frac{2x^2 - 8}{x^2 + x - 6} \quad \frac{2(x-2)(x+2)}{(x-2)(x+3)} \quad \frac{2(x+2)}{x+3}, x \neq 2$$

Ask: “**Are these the same function?**” After student responses, confirm they are. Then: “**Which form would you reach for first to find the zeros? The end behavior? The hole?**” This is the lesson’s central question: *three faces, three purposes*.

Lesson

Part 1 (10 min): Two Forms, Two Purposes.

Factored form $\Rightarrow$ structure at specific points (zeros, holes, VAs, domain). Standard form $\Rightarrow$ global behavior (degree, leading coeff, end behavior).

Give students the hook functions. Ask: “Which form reveals the hole at $x = 2$?” (Factored.) “Which reveals that the function approaches $+2$ as $x \rightarrow \infty$?” (Standard.)

Pose: what if we need the slant asymptote of $\frac{x^2 + x - 6}{x - 1}$? Neither form gives it directly — we need long division.

Part 2 (15 min): Polynomial Long Division.

Model $\frac{3x^3 - 5x^2 + 2x - 4}{x - 2}$ step by step on the board.

Emphasize: (1) descending order, (2) divide leading terms, (3) multiply and subtract, (4) repeat. Result: $q = 3x^2 + x + 4$, $r = 4$.

Slant asymptote rule: if $\deg(\text{num}) = \deg(\text{den}) + 1$, the slant asymptote is $y = q(x)$ (the remainder term $\rightarrow 0$).

Part 3 (10 min): Binomial Theorem.

Build Pascal’s Triangle through row 4 with the class. Expand $(x+2)^3$ using row 3: $x^3 + 6x^2 + 12x + 8$. Practice: $(x+3)^3$, then preview $(2x-1)^4$.

Active Monitoring

- Cold-call after Part 1: “Which form would you choose to find the domain restriction? Why?” Expect: factored form, because shared zeros are visible.
- During long division: watch for students who forget to subtract (sign errors) or who stop before $\deg(r) < \deg(g)$.
- Cold-call after Part 2: “If the quotient has degree 2, can it be a slant asymptote? Why not?” (Slant asymptote is linear.)
- During Pascal’s Triangle: confirm students know the row index matches the exponent, and watch for off-by-one errors.

Group Work & Differentiation

- **Tier R (Remediate):** Pre-factored numerator/denominator given; read zeros, holes, VAs directly; circle which form reveals end behavior.
- **Tier A (Apply):** Factor independently; apply multiplicity rule; execute long division on $g(x)$; determine horizontal asymptote from $h(x)$.
- **Tier E (Extend):** Long division to find slant asymptote; justify asymptote type; construct a rational function satisfying all given structural properties.

Individual Work & Assessment

Exit ticket (3 questions, 1 page): (1) identify which form reveals VAs/holes and state why; (2) complete the first step of long division for $\frac{x^2+3x+5}{x+2}$; (3) use Pascal’s Triangle (row 4) to expand $(x+2)^4$.

Collect tickets and sort: students who miss item 1 need re-teaching of form selection (LO 1.11.A); students who miss item 2 need scaffolded long division practice (LO 1.11.B); students who miss item 3 need Pascal’s Triangle reinforcement (LO 1.11.C).

Reinforcement & Extension

Homework (5 problems): form selection with two rational functions (zeros/holes/VAs and end behavior); full long division of a cubic by a linear; slant asymptote via long division including the remainder's meaning at an excluded value; binomial expansions using Pascal's Triangle including $(2x - 1)^4$; AP-style multiple choice on identifying holes via factoring ($x^3 - 8 = (x - 2)(x^2 + 2x + 4)$).

Extension: investigate why $\frac{x^3 - 8}{x - 2}$ has a hole (not a VA) by connecting the factored form to the division algorithm.

Preview of Lesson 1.12: transformations of polynomial and rational functions — how adding a constant inside or outside the function shifts the graph, connecting to the forms studied today.